

Theoretical Framework: From Local Trajectories to Global Latent Space

Let $\mathcal{X}$ denote the universal space of household observations. We partition this space into a finite set of discrete, non-overlapping geographies $g \in G$. To accommodate continuous dynamics, let time $\mathcal{T} \subset \mathbb{R}$ be a continuous interval, and $\mathcal{S} \subset \mathbb{R}^n$ denote the cross-sectional state space of household attributes.

The full set of observations for a specific geography is defined as the product space $X_g = \mathcal{T} \times \mathcal{S}$. We equip X_g with the standard Euclidean subspace topology. The universal space is the disjoint union $\mathcal{X} = \bigsqcup_{g \in G} X_g$, equipped with the disjoint union topology (meaning a subset is open in $\mathcal{X}$ if and only if its intersection with every geography X_g is open).

Let I be a set of indexed households. A specific household $i \in I$'s trajectory through time is given by a transport function $f_{t,k}^{g,i} : \mathcal{S} \rightarrow \mathcal{S}$, which maps household i 's realized state at time t to its own realized state at time k within the same geography.

We introduce an economic ordering $\succsim$ representing real wealth. Rather than assuming universal global comparability outright—which is fraught with spatial and intertemporal measurement issues—we impose comparability only along strictly defined, localized empirical dimensions. By explicitly leaving certain pairs incomparable, $\succsim$ constitutes an *incomplete* empirical preference relation.

Axiom 1 (Restricted Domain of Comparability). *Let $\succsim$ be an empirical binary relation representing real wealth. $\succsim$ is strictly defined only under the following localized conditions:*

- **Cross-Sectional Comparability:** Within any geography g and a fixed time $t \in \mathcal{T}$, any two household states $x = (t, s_x)$ and $y = (t, s_y)$ in X_g are directly comparable.
- **Trajectory Comparability:** For a specific household i 's trajectory, its state at time t , $x = (t, s) \in X_g$, and its transported state at time k , $x' = (k, f_{t,k}^{g,i}(s)) \in X_g$, are directly comparable.
- **Orthogonal Incomparability (Intertemporal Cross-Household):** The relation cannot be evaluated across both dimensions simultaneously in the empirical data. For distinct households $i \neq j$ observed at distinct times $t \neq k$, direct comparison is strictly undefined ($x_{i,t} \parallel x_{j,k}$).
- **Spatial Incomparability:** For any two observations in different geographies, $x \in X_g$ and $y \in X_h$ (where $g \neq h$), the relation is strictly undefined ($x \parallel y$).

Axiom 2 (Acyclicity and Continuous Transitive Closure). *Transitivity of the empirical relation $\succsim$ holds strictly within the defined domains in Axiom 1, but does not explicitly chain across orthogonal domains. However, the global relation $\succsim$ on X_g is strictly acyclic. Furthermore, we assume the transitive closure of this relation, denoted $\succsim^*$, forms a continuous partial preorder on X_g ,*

meaning its upper and lower contour sets are closed under the standard product topology of X_g .

Because we embrace the natural incompleteness of the empirical data, we do not require the topological space to be perfectly ordered. Instead, we demonstrate that a global machine learning architecture can validly map this disjoint and partially ordered space into a single continuous scalar dimension, preserving the known data while bridging the gaps via continuous spatial projection.

Lemma 1 (Local Continuous Representation). *For each geography $g \in G$, there exists a continuous function $u_g : X_g \rightarrow \mathbb{R}$ such that for all $x, y \in X_g$:*

$$x \succ^* y \implies u_g(x) > u_g(y) \quad \text{and} \quad x \sim^* y \implies u_g(x) = u_g(y),$$

where $\succ^*$ and $\sim^*$ denote the strict and symmetric parts of the transitive closure $\succsim^*$, respectively.

Proof. Fix an arbitrary geography $g \in G$. We verify the two hypotheses required to apply the continuous representation theorem for partial preorders (Herden, 1989; Ok, 2002).

Step 1: Second-countability of X_g . By definition, $X_g = \mathcal{T} \times \mathcal{S}$ where $\mathcal{T} \subset \mathbb{R}$ and $\mathcal{S} \subset \mathbb{R}^n$, so X_g embeds as a subspace of $\mathbb{R}^{1+n}$. The Euclidean space $\mathbb{R}^{1+n}$ is second-countable, admitting a countable base of open balls with rational centers and radii. Since second-countability is hereditary under subspaces, X_g is second-countable under the Euclidean subspace topology.

Step 2: $\succsim^*$ is a continuous partial preorder on X_g . We verify the three required properties.

Reflexivity. The base relation $\succsim$ is an economic ordering representing real wealth; any household state is at least as wealthy as itself, so $x \succsim x$ for all $x \in X_g$. This base reflexivity is inherited by $\succsim^*$.

Transitivity. Transitivity holds by the construction of the transitive closure.

Continuity. Axiom 2 stipulates that all upper contour sets $U(x) = \{y \in X_g : y \succsim^* x\}$ and lower contour sets $L(x) = \{y \in X_g : x \succsim^* y\}$ are closed in the product topology of X_g . This is precisely the definition of a *continuous* partial preorder.

Together, these three properties confirm that $(X_g, \succsim^*)$ is a continuous partial preorder. Note that completeness is not required.

Step 3: Existence of u_g . A continuous partial preorder with closed contour sets on a second-countable topological space admits a continuous, order-preserving scalar representation (Herden, 1989, Theorem 1; Ok, 2002, Theorem 1). Applying this result to $(X_g, \succsim^*)$ yields the existence of a continuous function $u_g : X_g \rightarrow \mathbb{R}$ satisfying the stated order-preservation conditions. Since $\succsim^*$ is the transitive closure of the base empirical relation $\succsim$, every comparison defined by $\succsim$ is preserved by $\succsim^*$, and hence by u_g . $\square$

Theorem 1 (Global Latent Representation). *If the incomplete relation $\succsim$ on the space $\mathcal{X}$ satisfies Axioms 1 and 2, there exists a global continuous scalar*

function $\hat{U} : \mathcal{X} \rightarrow \mathbb{R}$ providing a forward representation, such that for any empirically defined comparable pair $x, y \in X_g$:

$$x \succ y \implies \hat{U}(x) > \hat{U}(y) \quad \text{and} \quad x \sim y \implies \hat{U}(x) = \hat{U}(y).$$

Proof. By Lemma 1, for each geography $g \in G$ there exists a continuous, order-preserving local function $u_g : X_g \rightarrow \mathbb{R}$ representing $\succsim^*$ on X_g . Define $\hat{U} : \mathcal{X} \rightarrow \mathbb{R}$ piecewise by

$$\hat{U}(x) = u_g(x) \quad \text{where } x \in X_g.$$

Global continuity. The universal space $\mathcal{X} = \bigsqcup_{g \in G} X_g$ carries the disjoint union topology, under which a function is continuous if and only if its restriction to each component X_g is continuous. Since every u_g is continuous by Lemma 1, $\hat{U}$ is globally continuous on $\mathcal{X}$.

Order-preservation. Let $x, y \in X_g$ be any empirically comparable pair. Since $\succsim$ is defined within a single geography (Axiom 1) and u_g represents $\succsim^*$ on X_g , the stated forward implications hold immediately.

No violation of incomparability. By Axiom 1 (Spatial Incomparability), $\succsim$ defines no comparisons between X_g and X_h for $g \neq h$. Because $\hat{U}$ is only claimed to preserve comparisons defined within a single X_g , projecting disjoint geographies into the shared codomain $\mathbb{R}$ introduces no contradiction with any established economic relation. $\square$ $\square$

Remark on AI Latent Space and Residual Projections: Axiom 1 explicitly leaves the intertemporal relationship between distinct households, as well as cross-geography comparisons, formally undefined in the empirical data. However, a neural network maps these disjoint and partially ordered topologies into a shared, totally ordered latent space ($\mathbb{R}$).

Because $\mathbb{R}$ is totally ordered, for any two mathematically incomparable observations $x, y \in \mathcal{X}$, the model will inherently output scalar values such that either $\hat{U}(x) \geq \hat{U}(y)$ or $\hat{U}(y) \geq \hat{U}(x)$. These forced comparisons are **residual mathematical artifacts** of the latent projection (which is why Theorem 1 only claims a one-way, forward implication). They do not come from unfounded empirical assumptions forced onto the raw data, but rather emerge naturally and interpretably as the machine learning architecture interpolates across the shared continuous spatial and attribute features.

Furthermore, this topological formulation maintains a critical advantage for empirical measurement: it does not require the household physical transport function (f) to be continuous. Empirical data can be noisy and households can experience discontinuous shocks. As long as the underlying economic valuation of those states (Axiom 2) is continuous, a smooth latent space mapping remains mathematically valid and strictly preserves the true local intertemporal economic dynamics established in the data.